NFA, ε-NFA, Regular expressions

Question:
In each case below, find a string of minimum length in $\{a, b\}^*$ not in the language corresponding to the given regular expression.

a. $b^*(ab)^*a^*$
b. $(a^* + b^*)(a^* + b^*)(a^* + b^*)$
c. $a^*(baa^*)^*b^*$
d. $b^*(a + ba)^*b^*$

Question:
Consider the two regular expressions

$$r = a^* + b^*$$
$$s = ab^* + ba^* + b^*a + (a^*b)^*$$

a. Find a string corresponding to r but not to s.
b. Find a string corresponding to s but not to r.
c. Find a string corresponding to both r and s.
d. Find a string in $\{a, b\}^*$ corresponding to neither r nor s.

Question:
Let r and s be arbitrary regular expressions over the alphabet Σ. In each case below, find a simpler equivalent regular expression.

a. $r(r^*r + r^*) + r^*$
b. $(r + \varepsilon)^*$
c. $(r + s)^*rs(r + s)^* + s^*r^*$

Question:
In each case below, give a simple description of the smallest set of languages that contains all the "basic" languages $\emptyset$, $\{\varepsilon\}$, and $\{\sigma\}$ (for every $\sigma \in \Sigma$) and is closed under the specified operations.

a. union
b. concatenation
c. union and concatenation

Question:
Suppose w and z are strings in $\{a, b\}^*$. Find regular expressions corresponding to each of the languages defined recursively below.

a. $\varepsilon \in L$; for every $x \in L$, then wx and xz are elements of L.
b. $a \in L$; for every $x \in L$, wx, xw, and xz are elements of L.
c. $\varepsilon \in L$; $a \in L$; for every $x \in L$, wx and zx are in L.

Question:
Find a regular expression corresponding to each of the following subsets of $\{a, b\}^*$.

a. The language of all strings containing exactly two a's.
b. The language of all strings containing at least two a's.
c. The language of all strings that do not end with ab.
d. The language of all strings that begin or end with aa or bb.
e. The language of all strings not containing the substring aa.
f. The language of all strings in which the number of a's is even.

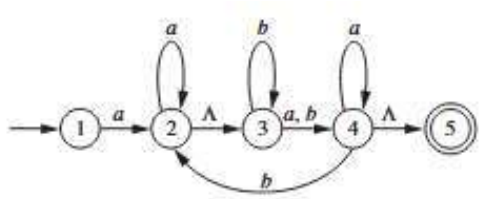
g. The language of all strings containing no more than one occurrence of the string aa. (The string aaa should be viewed as containing two occurrences of aa.)
h. The language of all strings in which every a is followed immediately by bb.
i. The language of all strings containing both bb and aba as substrings.
j. The language of all strings not containing the substring aaa.
k. The language of all strings not containing the substring bba.
l. The language of all strings containing both bab and aba as substrings.
m. The language of all strings in which the number of a's is even and the number of b's is odd.
n. The language of all strings in which both the number of a's and the number of b's are odd.

Question:
The regular expression $(b + ab)*(a + ab)*$ describes the set of all strings in $\{a, b\}*$ not containing the substring _____x_____ for any x. (Fill in the blanks appropriately.)

Question:
The regular expression $(a + b)*(aa*bb*aa* + bb*aa*bb*)(a + b)*$ describes the set of all strings in $\{a, b\}*$ containing both the substrings _____ and _____ .

Question:
The star height of a regular expression r over Σ, denoted by sh(r), is defined as follows:
i. $sh(\emptyset) = 0$.
ii. $sh(\varepsilon) = 0$.
iii. $sh(\sigma) = 0$ for every $\sigma \in \Sigma$.
iv. $sh((rs)) = sh((r + s)) = \max(sh(r), sh(s))$.
v. $sh((r*)) = sh(r) + 1$.
Find the star heights of the following regular expressions.
a. $(a(a + a*aa) + aaa)*$
b. $(((a + a*aa)aa)* + aaaaaa*)*$

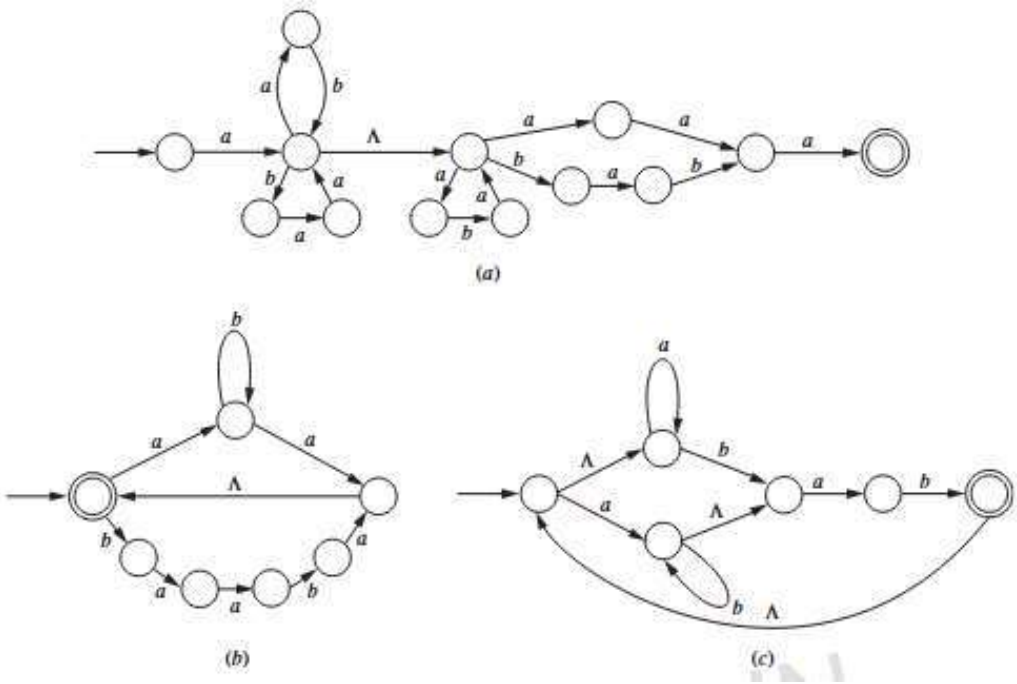
Question:
For the NFA in the below Figure,



Construct the transition diagram. For each string below, say whether the NFA accepts it.
a. aba
b. abab
c. aaabbb

Find a regular expression corresponding to the language accepted by the NFA pictured in Figure

Question:
For each of the NFAs shown in below Figure, find a regular expression corresponding to the language it accepts.



Question:
Below is the transition table for an NFA with states 1–5 and input alphabet {a, b}. There are no ε-transitions.

q	$\delta(q, a)$	$\delta(q, b)$
1	{1, 2}	{1}
2	{3}	{3}
3	{4}	{4}
4	{5}	Ø
5	Ø	{5}

a. Draw a transition diagram.
b. Calculate δ∗(1, ab).
c. Calculate δ∗(1, abaab).

Question:
A transition table is given for an NFA with seven states.

q	$\delta(q, a)$	$\delta(q, b)$	$\delta(q, \Lambda)$
1	Ø	Ø	{2}
2	{3}	Ø	{5}
3	Ø	{4}	Ø
4	{4}	Ø	{1}
5	Ø	{6, 7}	Ø
6	{5}	Ø	Ø
7	Ø	Ø	{1}

Find:
a. ε({2, 3})

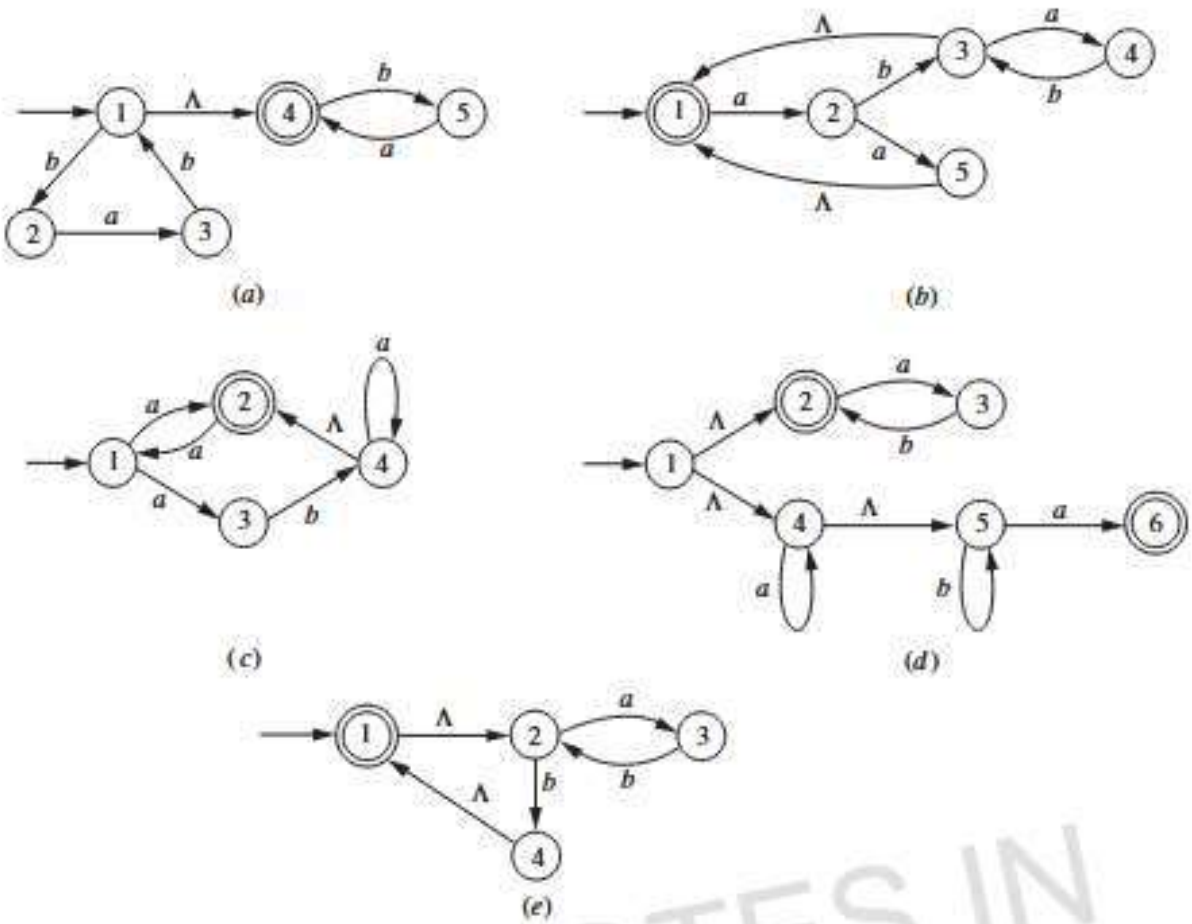
b. ε({1})
c. ε({3, 4})
d. δ∗(1, ba)
e. δ∗(1, ab)
f. δ∗(1, ababa)

Question:
A transition table is given for another NFA with seven states.

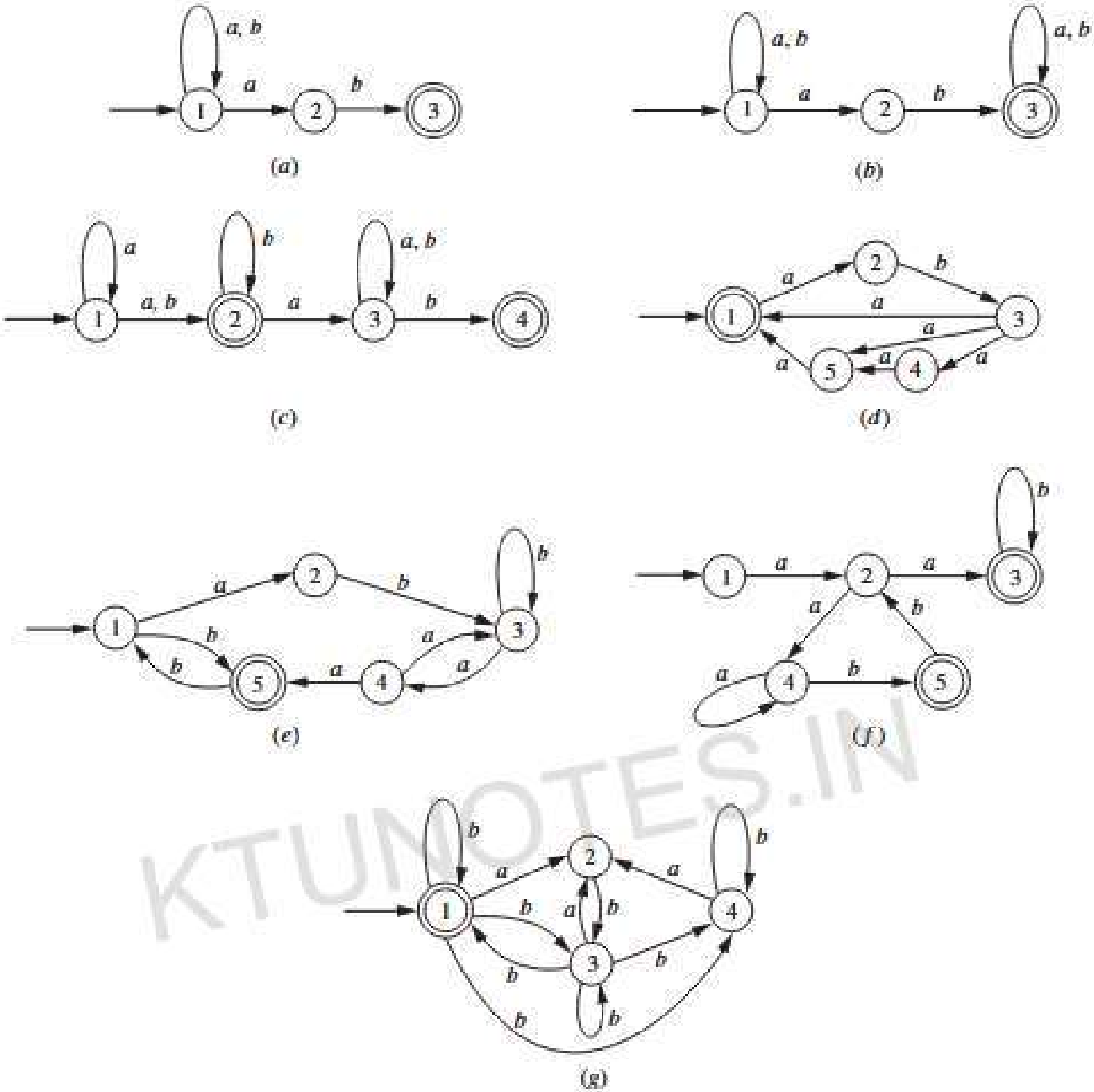
q	$\delta(q, a)$	$\delta(q, b)$	$\delta(q, \Lambda)$
1	{5}	∅	{4}
2	{1}	∅	∅
3	∅	{2}	∅
4	∅	{7}	{3}
5	∅	∅	{1}
6	∅	{5}	{4}
7	{6}	∅	∅

Calculate δ∗(1, ba).

Question:
Consider the NFAs in below figure. Use the algorithm to draw NFA with no ε-transitions, accepting the same language.



Question:
For the NFAs in the below figure, using the subset construction, draw an FA accepting the same language.



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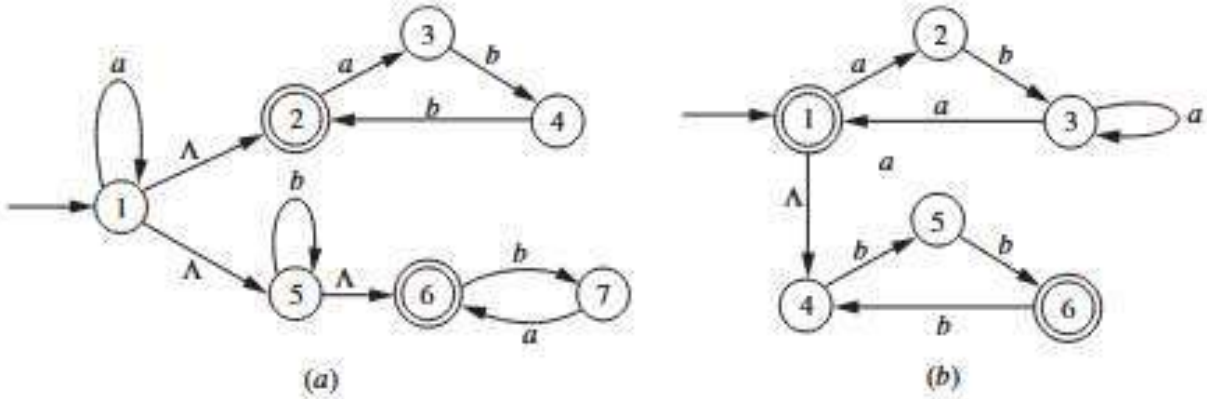
Question:
Suppose L ⊆ ε∗ is a regular language. If every FA accepting L has at least n states, then every NFA accepting L has at least _____states. (Fill in the blank, and explain your answer.)

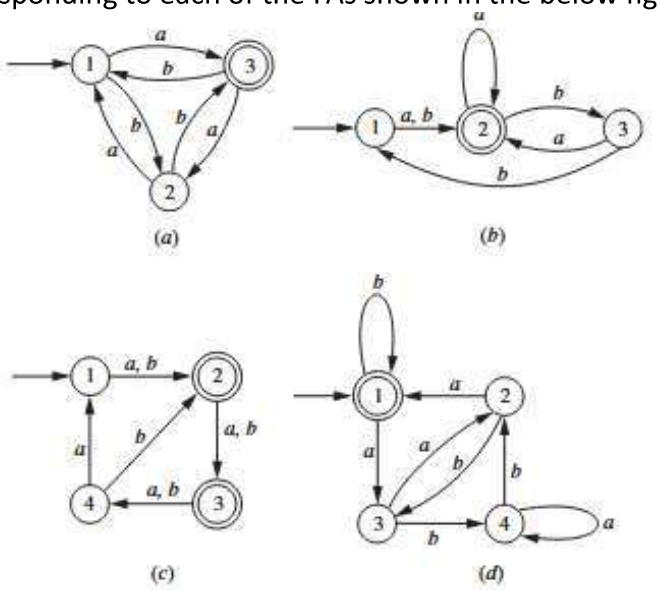
Question:
Consider the NFA in the figure below. Draw an FA accepting the same language.



Question:
For each of the following regular expressions, draw an NFA accepting the corresponding language, so that there is a recognizable correspondence between the regular expression and the transition diagram.

a. (b + bba)∗a
b. (a + b)∗(abb + ababa)(a + b)∗
c. (a + b)(ab)∗(abb)∗
d. (a + b)∗(abba∗ + (ab)∗ba)
e. (a∗bb)∗ + bb∗a∗

Question:
Below figure shows FAs M1 and M2 accepting languages L1 and L2 , respectively. Draw NFAs accepting each of the following languages, using the constructions.

a. L2* ∪ L1
b. L2 L1∗
c. L1L2 ∪ (L2L1)∗

Question:
Find a regular expression corresponding to each of the FAs shown in the below figure.